

secures the connector to remain intact and engaged, and terminal position assurance which prevents the terminal from backing out while ensuring that contacts remain fully seated in the connector.

High-power connectors combine high conductivity and ultra low resistance. They are provided with air-cooling mechanisms to reduce heating effects. The increasing usage of optical fibre connectors in high-bandwidth applications is now demanding newer and quicker termination technologies.

Connectors going contactless. An interesting technology that came up in the connector space in the last two years was the concept of contactless connectivity. This technology meets the market need of miniaturisation, decentralisation and the need for smarter systems. It allows you to transfer power and data without signal loss or the need for mechanical coupling through messy connection cables. It can operate through water, oil and non-metallic barriers, and is immune to pollution, vibration and harsh environment. It has low maintenance cost.

Hybrid technology. Hybrid connectors combine both power and data signal elements into a single housing. You can replace them with a single hybrid connector unlike traditional designs where you have to use separate data and power connections. They are widely used outdoors in heated and dusty environments and also under special IP compliances such as IP65 and IP66.

Hybrid connectors are commonly used in avionics and also in video recorders for security surveillance. Combining power and data communication on a single connector helps reduce the number of connector elements, bill of materials (BOMs) and the size of the entire system, though this is subjective—varies for each case.

Hermaphrodites and stacking connectors. Conventional connectors are of two varieties: male-ended and female-ended. A female-ended connector cannot be replaced with a male-ended connector and vice versa.

Operating Conditions for Sealed Rugged Connectors

Criteria	Condition
Substances to be blocked by sealing	Water or dust
Temperature range	For normal use, -20°C to +60°C For military use, extreme -50°C to +150°C
Pressure differential	Less than 0.2 bar
Type of exposure	Most common is a splash type, but jets or immersion are also possible

In the case of new hermaphroditic or genderless connector, it has mating surfaces that include both male and female aspects simultaneously, that is, it has a 'mate to itself' design. This feature removes the gender constraints and allows these components to freely connect with others, provided they are of similar size and type. The hermaphrodite connectors are also informally called as androgynous, sexless, combination, combo, two-in-one and two-way.

The stackable connector systems have an array of male pins on one surface and a complementary array of female sockets on the opposite surface. This allows stacking of multiple units.

Factors to consider for selecting the best connector

Connectors for embedded systems can be of two types based on the application or the type of transmission they allow—power or data. You have to know whether they would be used on a printed circuit board (PCB) or in a harness assembly. Another typical division is based on whether they are input connectors, on-board connectors or output connectors. The main consideration for choosing a connector would be the end application of the embedded computer.

There are multiple criteria based on which a connector can be chosen depending upon its application.

In case of the input and output (I/O) connectors, the main parameters to look at are size and number of I/O signals required for your design,

amongst other things.

For coaxial radio frequency (RF) connectors, the frequency range and attenuation levels are what matter the most.

For power connectors, the factors that have more emphasis are voltage ranges, current ranges and operating temperatures the connector can withstand.

The connectors used in aerospace and defence industries require safety, security and ability to withstand harsh environments. More rugged types of connectors, which cannot be easily bypassed, are the kind required here. Other properties, such as material science, also vary depending upon several technical and commercial factors.

Compactness and portability. The size and weight of a device can be an important factor in certain application fields. If the dimensions of the product forbid the designers to use a particular type of connector, they might have to go for a non-standard way of using different interfaces on it.

Miniaturisation and portability requirements are very high on consumer electronics end. The size of PCB is becoming smaller and the design of connector is also changing accordingly.

Another area where compactness and portability issues come up is the avionics sector. On devices such as unmanned aerial vehicles (UAVs), vibration form factor, the size, height and weight are very important criteria. The overall electronics used in UAVs are bound to be very small in size, and that is when portability of connector becomes a key aspect.

The industrial requirement for embedded systems would not maintain so

Scalability

A scalable electronics system is the one that can adapt to increasing workloads by expanding the hardware, and the performance of the device increases in proportion to it. A connector with high scalability can lower the hardware development cost and time to market.